University of Pisa

Sant'Anna School of Advanced Studies

Master's Degree in Economics

Ecological inference and US elections

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1 Introduction

Introduction: Kuriwaki and McCartan (2026), but also McCartan and Kuriwaki (2025).

I should say something about the more general importance of estimating correctly when we do not observe individual-level data.

I should say that sometimes we do not observe individual-level data and we might be tempted to infere something based on highly granular data. But these inferences may be wrong.

That could be relevant for economists in various applications and analyses and it is not solely a nuisance for econometricians.

In fact, data are usually reported by official bureau of statistics even at highly granular level but not at individual level and some relationships become masked.

Subsequently, I should present the problem with appropriate notation.

And probably I should also mark the difference between this problem and others problem of aggregation such as the one discussed in BLP model.

In any case, I will later decide how to structure the dissertation and where to put the complete explanation of the method: in principle, I would put it into the introduction and later I would present the syntethic example just referring to what I have already written here.

2 A synthetic example

In this section I will show a synthetic example that helps explaining why the problem is relevant and why the solution adopted by McCartan and Kuriwaki (2025) helps us in solving it better than naive methods.

In the synthetic example we are the ones who generate the data and therefore we know the exact model of the reality. We show that, if we generate the data and then we try to estimate with different methods we obtain different results and the naive methods do not work well.

In particular, we can exploit the constitutional referendum in Italy to show a possible way in which the naive estimators could not work and therefore we can generate a dataset that encompasses the problem.

The problem is the following: we want to estimate how did voters with different levels of income vote.

If we look at data at regional level, we would be tempted to say that richest people voted more in favour of the government proposal. In fact, the three regions where the votes in favor were more than the votes in contrast (Lombardy, Veneto and Friuli-Venezia Giulia) are among the richest regions in Italy. On the contrary, the regions were No got more than 60% are among the poorest regions of Italy (Basilicata, Campania, Sicily).

However, once we look at data at municipal level, we find that, within the regions, the relationship is nearly inverse: No votes were more frequent in urban areas and other municipalities where income is relatively higher.

That divergence makes us think that if we were able to observe results at individual level, we could find even different relationships between income and voting behavior.

Therefore, we can now generate a synthetic dataset that presents a structure that resembles some stylized facts that we observe in the constitutional referendum results and other notions that we have about electoral behaviors in Italy.

Our synthetic experiment is constructed as follows.

We assume that election takes place in a Country which has three different regions: North, Centre and South.

Each Country is divided in 100 precincts, so that in the end we observe 300 precincts: 100 precincts in the North region, 100 in the Centre region, 100 in the South region.

For each electoral precinct, we observe the percentage of high-income and low-income voters and the electoral results.

Within each income category, there is high variability in their electoral behaviour.

In particular, we impose that high-income people in the North and South precincts have a high propensity of voting in favor of the reform, while in the Centre this percentage is much lower. As for low-income people, we impose that in North they have high propensity of voting in favor, while in the Centre and in the South this propensity is much lower.

At the same time, we impose that the percentage of high-income voters is highest in the North and lowest in the South.

3 An application to US data

An application to US data

4 Other Comments

Other comments

5 Conclusions

Conclusions

6 References

References

Kuriwaki, Shiro and McCartan, Cory (2026), *The Role of Confounders and Linearity in Ecological Inference: A Reassessment*, arXiv: 2601.07668 [stat.AP], URL: https://arxiv.org/abs/2601.07668.

McCartan, Cory and Kuriwaki, Shiro (2025), *Identification and Semiparametric Estimation of Conditional Means from Aggregate Data*, arXiv: 2509.20194 [stat.ME], URL: https://arxiv.org/abs/2509.20194.